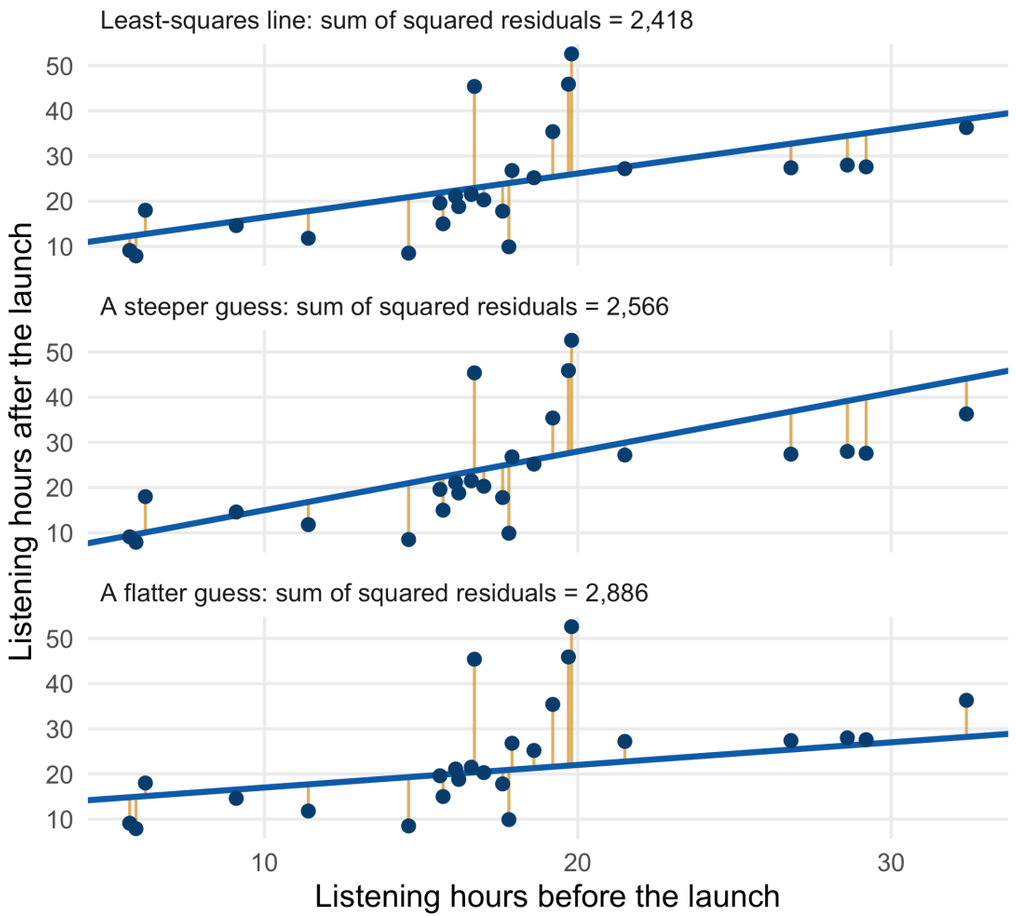
Which line? Least squares



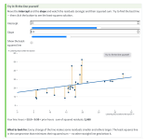
- Residual = observed − predicted (orange segments)
- **OLS** = the line that **minimizes the sum of squared residuals**:

$$\min_{\hat{\beta_0},\hat{\beta_1}} \sum_{i=1}^{n} \left(y_i - \hat{\beta_0} - \hat{\beta_1} x_i\right)^2$$

- Squaring: big misses count a lot; above and below alike · `lm()` does exactly this

Live demo: fit the line yourself — course book, §5.2

https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-line



Marketing Analytics

Chapter 5 · Learning from Observational Data

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November 2026

Part 1 · Observational data I

Before we start

Paper quiz on Chapter 4 *(10 min)*

- Individual, closed book, counts toward participation
- Afterwards: a few leftovers from Chapter 4 and a look at the quiz questions

Assignment 2 is running: deadline Group A Sun, Nov 8 · Group B Wed, Nov 11; debrief at the start of next session.

Devices only when we code

- **Input blocks:** laptops closed, phones away
- **Coding blocks** (2 × 25 min): laptops open, alone or with your neighbor; one group presents at the end

> Today: correlation and the regression line → **coding block 1** → adjustment and exogeneity → **coding block 2** → prediction vs. explanation.

Today’s session

Observational data I *(today)*

5.1 Smart Mix without randomization
5.2 Correlation, the line, and least squares
5.3 Reading a regression
5.4 Adjusting for confounders — and exogeneity
5.5 Categorical predictors
5.6 Prediction vs. explanation

Observational data II *(next session)*

5.7 Functional form and elasticities
5.8 Logistic regression
5.9 Diagnostics that change decisions
5.10 Difference-in-differences

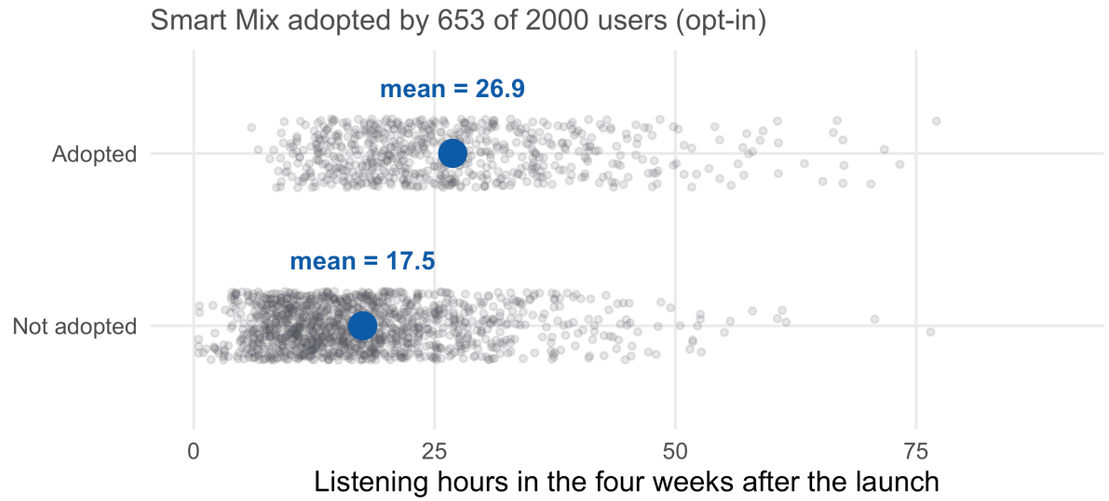
> **Driving question:** *What if we cannot randomize?*

5.1 Smart Mix without randomization

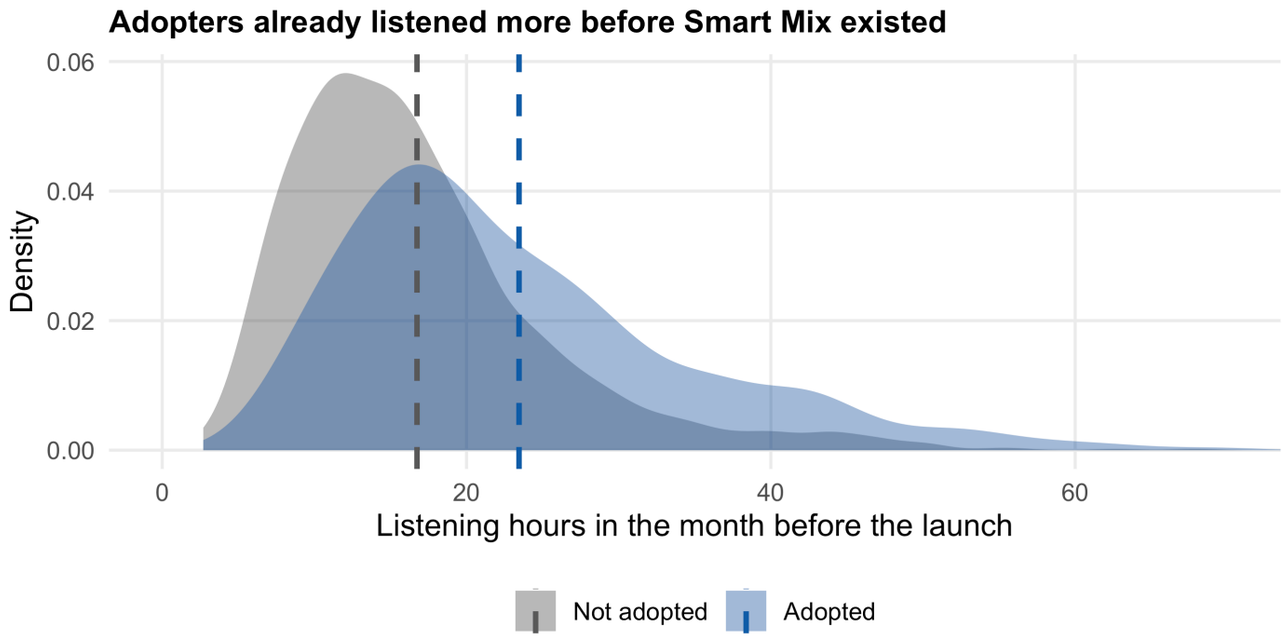
A different history

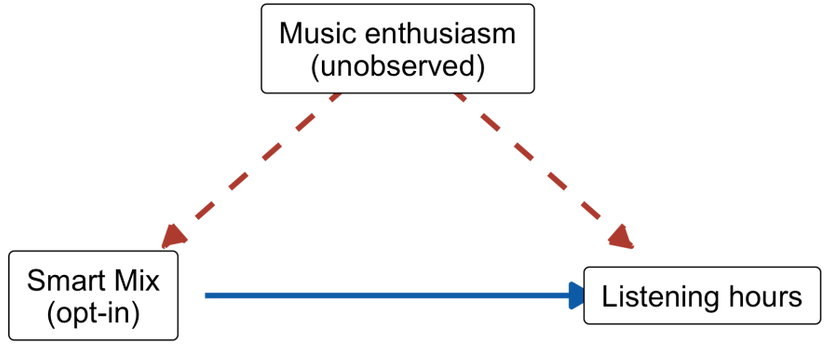
- Chapter 4: A/B test, 400 users → **+3.2 hours**
- Now: no experiment. Smart Mix launched as an **opt-in** feature
- Four weeks later: data of **2,000 users**, 653 adopters

> The dashboard says: adopters listen **9.3 hours more** — three times the experimental effect. Would you report this number to management?



Adopters were different before Smart Mix existed

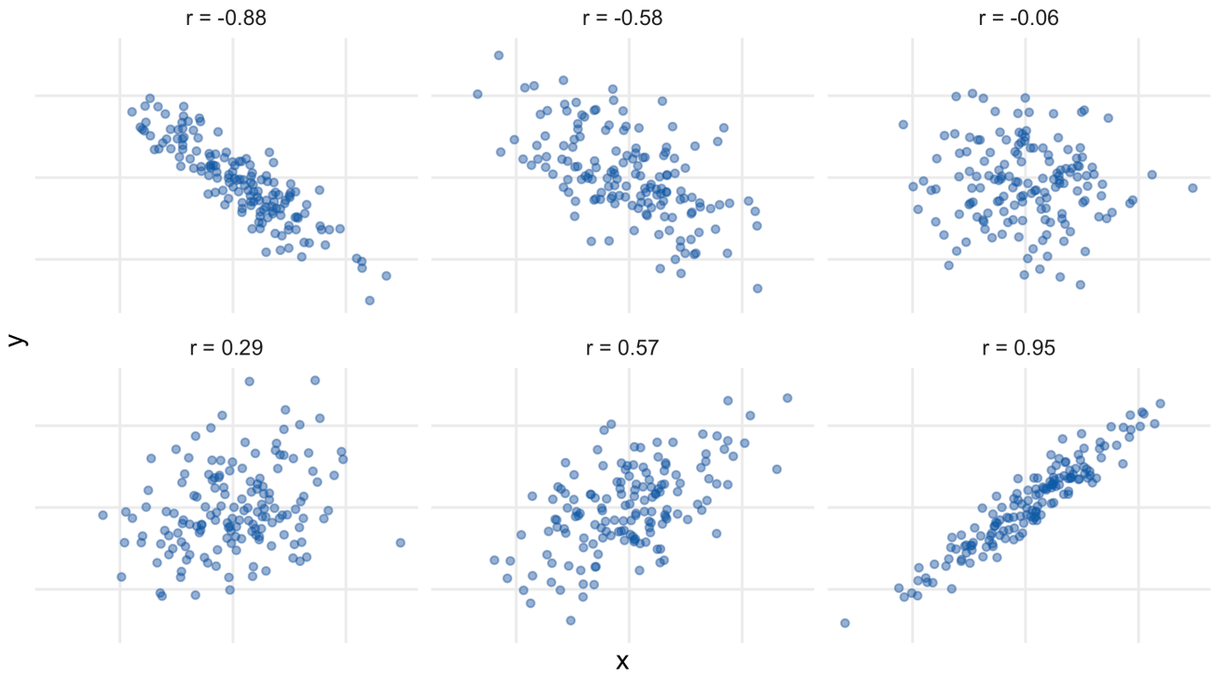




Same costume, new villain: **music enthusiasm** drives adoption *and* listening. Without a coin flip, we have one tool left: **adjustment**.

5.2 From correlation to a line

The correlation coefficient

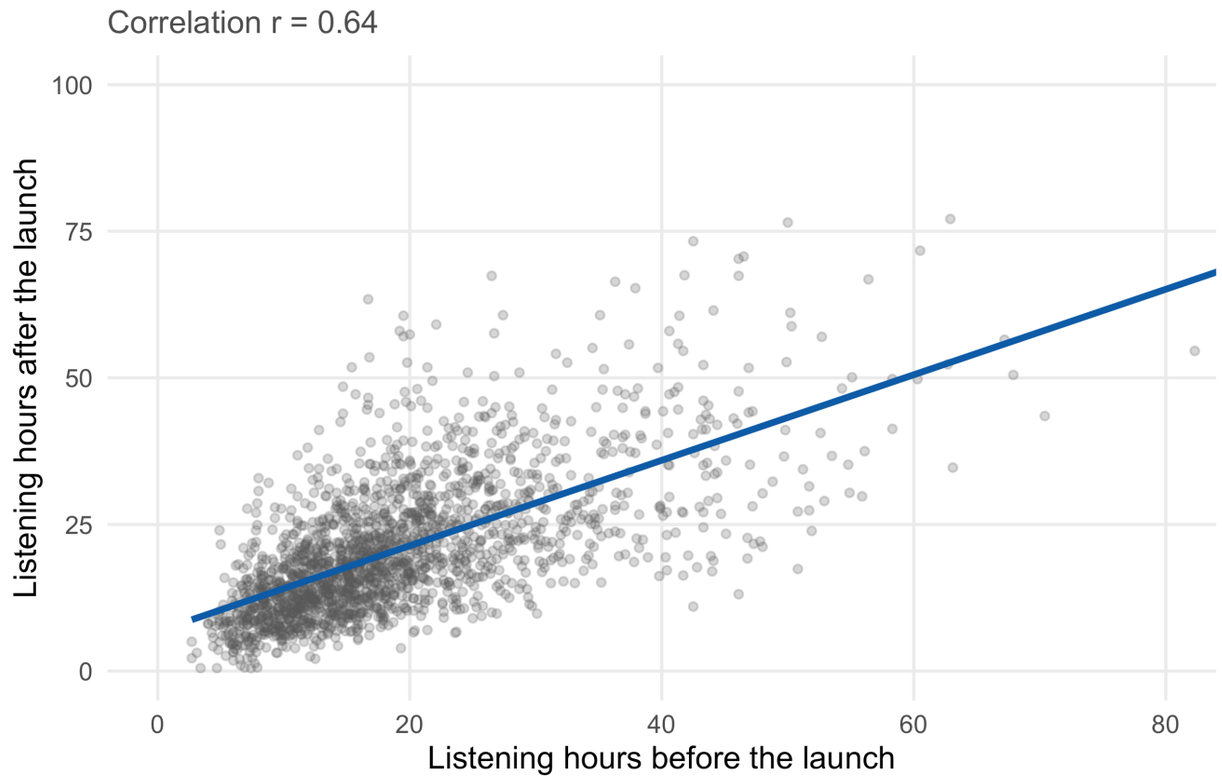


$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}$$

- Standardized covariance → always between **−1 and +1**, no units
- **Sign** = direction · **size** = how tightly the cloud hugs a straight line
- Rules of thumb: 0.1 small · 0.3 medium · 0.5 large
- Smart Mix: `cor(prior_hours, hours)` = **0.64**

Two caveats: *r* measures **linear** association only (a U-shape can have $r \approx 0$) — and it is **symmetric**: nothing about which variable causes which.

Two variables, one line



$$\text{hours}_i = \beta_0 + \beta_1 \text{ prior hours}_i + \varepsilon_i$$

- i = one user · β_0 **intercept** · β_1 **slope** · ε_i **error term**
- Correlation: *that* they move together · regression: *how much*

```
1 coef(lm(hours ~ prior_hours, data = smartmix_obs))
```

```
(Intercept) prior_hours
     6.7849      0.7291
```

> Each extra hour before the launch **goes with** 0.73 hours after it — *goes with*, not *causes*.

5.3 Reading a regression

The full output

```r
model_naive <- lm(hours ~ smart_mix, data = smartmix_obs)
summary(model_naive)
```

```
Call:
lm(formula = hours ~ smart_mix, data = smartmix_obs)

Residuals:
   Min     1Q Median     3Q    Max
-20.88  -7.64  -2.06   4.76 101.22

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   17.541      0.300    58.5   <2e-16 ***
smart_mix      9.341      0.525    17.8   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

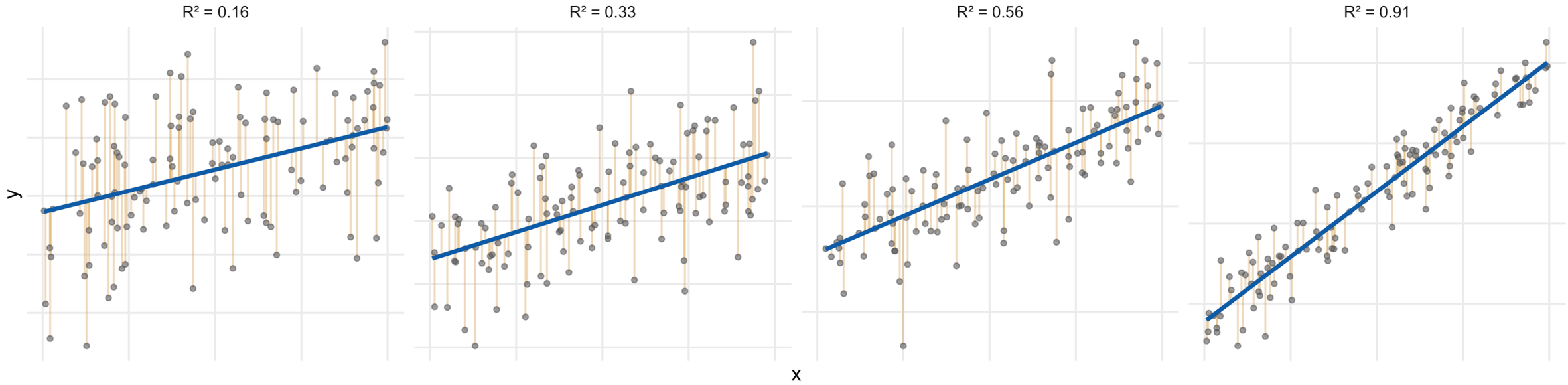
Residual standard error: 11 on 1998 degrees of freedom
Multiple R-squared:  0.137, Adjusted R-squared:  0.136
F-statistic:  316 on 1 and 1998 DF,  p-value: <2e-16
```

Line by line

Output	Meaning
`Estimate`	Intercept = mean of non-adopters (18.4); `smart_mix` = difference adopters − non-adopters (9.3)
`Std. Error`	Uncertainty of each coefficient (Chapter 3)
`t value`, `Pr(>|t|)`	*t*-test of each coefficient against zero (Chapter 4): signal / noise
`Residual standard error`	Typical distance of a user from the line, in hours
`Multiple R-squared`	Share of the variation in hours explained by the model (14%)
`F-statistic`	The whole model against “no predictors”

> With one 0/1 predictor, the regression **is** the comparison of two means — and its *t*-test is the two-sample *t*-test. `confint(model_naive)` gives the confidence intervals.

R^2: the share of explained variation



- Same idea as η^2 in the ANOVA: $R^2 = SS_M / SS_T$ — the share of the variation in y that the line explains
- Smart Mix dummy: 0.14 · prior listening: 0.41

> R^2 says how well the line **fits** — nothing about whether the slope is **causal**, and a small R^2 does not make an effect unimportant (14% explained, but 9 hours).

Coding block 1: podcast promotion (25 min)

The case. A podcast network tracks 400 episodes: promotion spend (euros), the host's followers, and downloads in the release week.

Data: `podcast_promo.csv` — section "Coding block 1" in `05-observational-data.R`

Laptops open — alone or with your neighbor.

Tasks

1. **Scatter plot** of downloads (y) against promotion spend (x); compute the **correlation**; one sentence
2. Fit the **regression** with `lm()`, look at `summary()`: interpret the slope in words; read off R^2
3. Is the slope a **causal** effect of promotion? What would you need to argue?

> At the end: one group presents its numbers.

5.4 Adjusting for confounders

Holding prior listening constant

```
model_prior <- lm(hours ~ smart_mix + prior_hours, data = smartmix_obs)
round(coef(summary(model_prior)), 3)
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    6.451      0.412   15.65        0
smart_mix      4.901      0.443   11.07        0
prior_hours    0.662      0.020   33.17        0
```

> The Smart Mix coefficient drops from **9.3** to **4.9**: now the difference between adopters and non-adopters **with the same prior listening**. Part of the naive gap belonged to prior listening — and is now attributed to it.

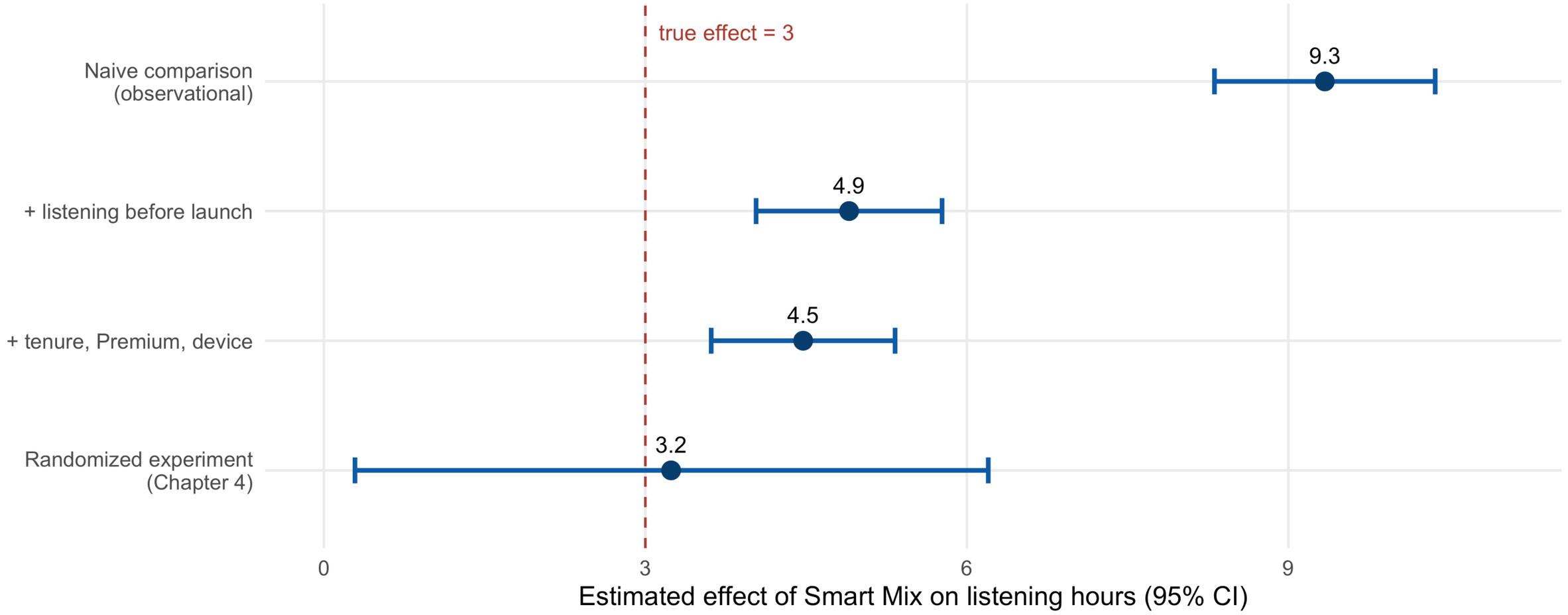
More variables, same logic

```
model_full <- lm(hours ~ smart_mix + prior_hours + tenure + premium + device, data = smartmix_obs)
round(coef(summary(model_full)), 3)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)     6.255      0.540  11.575    0.000
smart_mix       4.472      0.437  10.223    0.000
prior_hours     0.613      0.021  29.688    0.000
tenure          0.038      0.011   3.344    0.001
premium         2.924      0.426   6.869    0.000
deviceAndroid  -1.475      0.422  -3.493    0.000
deviceDesktop  -2.488      0.576  -4.320    0.000
```

> Each coefficient: the association with the outcome **holding all other variables in the model constant**. Tenure, Premium and device change little: ≈ **4.5 hours**.

Naive, adjusted — and the truth

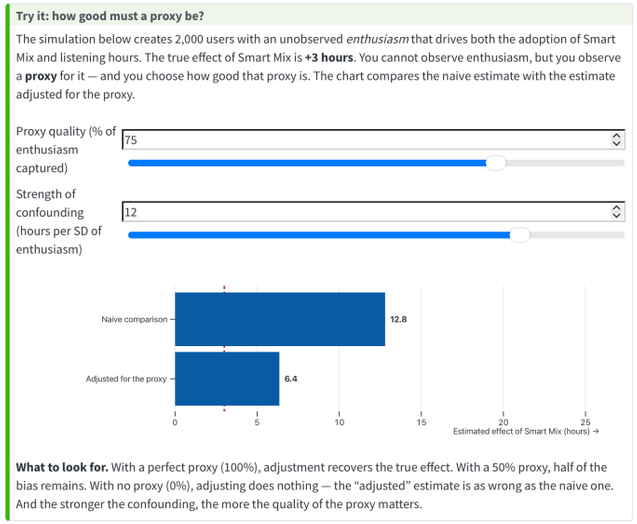


Adjustment removed ~75% of the bias — **not all of it**. We adjusted for a *proxy* of enthusiasm, not for enthusiasm. Regression removes exactly the confounding your variables capture.

Live demo: how good must a proxy be?

Course book, §5.4 — set the **quality of the proxy** and the **strength of confounding**.

https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-adjust



With a 50% proxy, how much of the bias is left? And what happens at 0%?

Two warnings

Adjusting for the wrong variable makes it worse. A **mediator** (on the causal path) absorbs part of the effect you want. Only **confounders** — determined *before* the treatment, affecting treatment *and* outcome — belong in the model.

"Enough" is untestable. The assumption of no unobserved confounders cannot be checked in the data. Report *what* you adjusted for — and what you plausibly could not.

The assumption behind adjustment: exogeneity

$$\text{hours}_i = \beta_0 + \beta_1 \text{ smart mix}_i + \beta_2 \text{ prior hours}_i + \cdots + \varepsilon_i$$

- ε_i = everything the model leaves out: mood, taste, time — and **enthusiasm**
- β_1 is the causal effect only if $\text{Cov}(\text{smart mix}_i, \varepsilon_i) = 0$: **exogeneity**
- Randomization guarantees it. Opt-in violates it: enthusiasm drives adoption **and** sits in ε_i → Smart Mix is **endogenous**
- Each adjustment variable moves a piece of ε_i into the model: `prior_hours` took most of the enthusiasm out (9.3 → 4.5); what is left still correlates with adoption (4.5 > 3)

Exogeneity cannot be tested. The residuals of a fitted regression are uncorrelated with the predictors *by construction* — the data always "look" exogenous. The argument has to come from outside: theory, and how the data came to be (who chose the treatment, and why).

Your turn (5 min)

Add **age** to the full model. Does the Smart Mix coefficient change? Why (not)?

```
model_age <- lm(hours ~ smart_mix + prior_hours + tenure + premium + device + age,
                data = smartmix_obs)
round(coef(summary(model_age)), 3)
```

Then: which variable in the data set should you **never** adjust for when estimating the effect of Smart Mix on listening — and why?

5.5 Categorical predictors

Dummies and the reference category

```
coef(model_full)[c("premium", "deviceAndroid", "deviceDesktop")]
```

```
premium deviceAndroid deviceDesktop
  2.924        −1.475        −2.488
```

- **0/1 variable** (`premium`): coefficient = difference between the two groups, other variables held constant
- **Several categories** (`device`): R picks a **reference** (first level: iOS) and creates one **dummy** per other category
- Each dummy = difference to the **reference**

> Android users: **1.5 hours less** than iOS users; Desktop: **2.5 hours less** — holding everything else constant. *p*-values compare each category with the reference; “does device matter at all?” → `anova(model_without, model_full)` (the ANOVA of Chapter 4).

Coding block 2: a loyalty program (25 min)

The case. An online shop lets customers **opt into** its loyalty program. 1,500 customers: membership (1/0), spending this year, spending last year, tenure, device.

Data: `loyalty_program.csv` — section “Coding block 2” in `05-observational-data.R`

Laptops open — alone or with your neighbor.

Tasks

1. **Naive:** regress `spending` on `member`; interpret — and why not causal?
2. **Adjust** for `spend_prev`, `tenure`, `device`: how does the coefficient change, what does it mean now? Report with 95% CI
3. Interpret the coefficient of `device`
4. Is membership **exogenous** after adjustment? Argue from the data-generating process

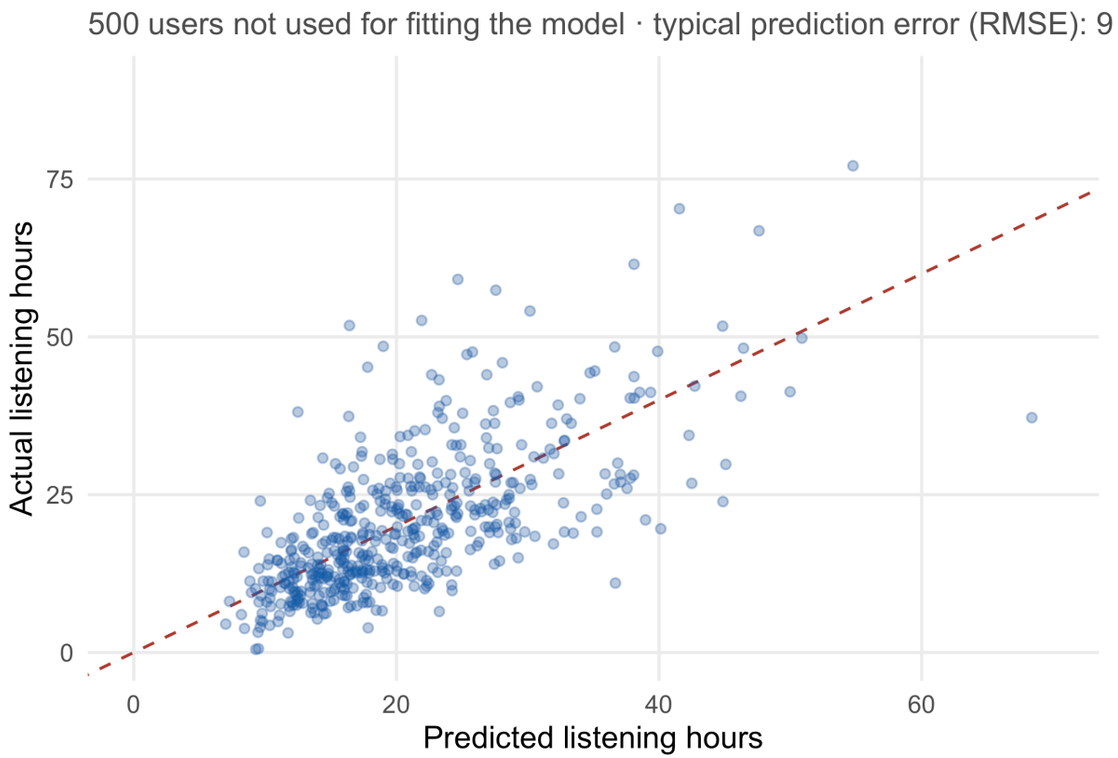
> At the end: one group presents its numbers.

5.6 Prediction vs. explanation

Predicting listening time for new users

```r
set.seed(42)
train_ids <- sample(nrow(smartmix_obs), 1500)
train <- smartmix_obs[train_ids, ]
test  <- smartmix_obs[-train_ids, ]
model_pred <- lm(hours ~ prior_hours + tenure + premium +
                   device + smart_mix, data = train)
test$predicted <- predict(model_pred, newdata = test)
sqrt(mean((test$hours - test$predicted)^2))   # RMSE
```

```
[1] 9.425
```



For **prediction**, it does not matter *why* a variable is associated with the outcome — only how well the predictions hold up on **data the model has not seen**.

Two questions, two rules

Explanation (causal question): include confounders, exclude mediators, state what is unobserved. The **coefficient** matters; R^2 is secondary.

Prediction: include whatever predicts; judge by **out-of-sample errors**. Coefficients are not interpreted; the **predictions** matter.

A model can be a **good predictor and a bad explainer** at the same time. Where in your company would each mistake — reading causal effects off a prediction model, or judging a causal model by its R^2 — cost money?

Learning check (Part I)

1. Slope of hours on prior hours = 0.73. Interpret — and why is it not causal?
2. Adding a variable can increase, decrease, or leave the Smart Mix coefficient unchanged. When does each happen?
3. `planFamily` = 4.2 with reference “Free”. Interpret. How do you test whether the plan matters at all?
4. R^2 = 0.71 for predicting next month’s spending with 25 predictors on 300 customers. What do you ask?

Next session

Observational data II — beyond the straight line, and a quasi-experiment

- Functional form: when a line is wrong — **elasticities**
- Yes/no outcomes: logistic regression
- The three diagnostics that change decisions
- **Difference-in-differences** with real playlist data

> Before next session: work through §5.1–5.6 on the course site. **Debrief of Assignment 2** at the start of the session.

Part 2 · Observational data II

Before we start

Debrief A2 + quiz Chapter 5, Part I *(15 min, one sheet)*

- Page 1: debrief of **Assignment 2** (3 assignment points)
- Page 2: quiz on **Chapter 5, §5.1–5.6** (participation)
- Closed book, no laptop

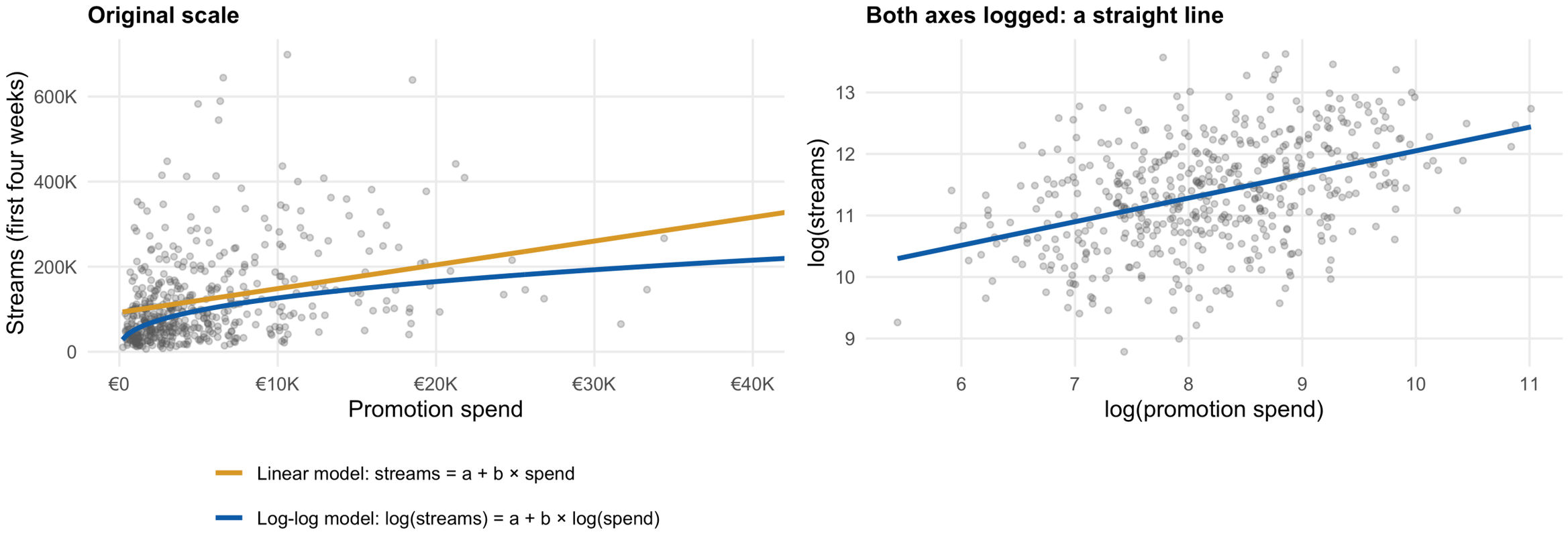
Assignment 3 is handed out today (end of session).

Devices only when we code — as last time

> Today: functional form and elasticities → **coding block 3** → logistic regression, diagnostics → **coding block 4** → difference-in-differences → A3.

5.7 Functional form: when the line is wrong

Diminishing returns



500 promotion campaigns: the first euros buy many streams, later euros fewer. A straight line cannot bend — on **log scales**, the cloud becomes a line again.

The multiplicative model — and why logs make it linear

Multiplicative: each predictor *scales* the outcome

$$\text{streams}_i = e^{\beta_0} \cdot \text{spend}_i^{\beta_1} \cdot \text{followers}_i^{\beta_2} \cdot e^{\varepsilon_i}$$

Take logs on both sides → additive again, a job for `lm()`:

$$\log(\text{streams}_i) = \beta_0 + \beta_1 \log(\text{spend}_i) + \beta_2 \log(\text{followers}_i) + \varepsilon_i$$

Three log rules do the work:

$\log(a \cdot b) = \log a + \log b$ $\log(a^b) = b \log a$

$\log(e^x) = x$

Additive model: each euro adds β_1 streams, always.
Multiplicative model: each euro adds a **percentage** of the current streams — diminishing returns built in.

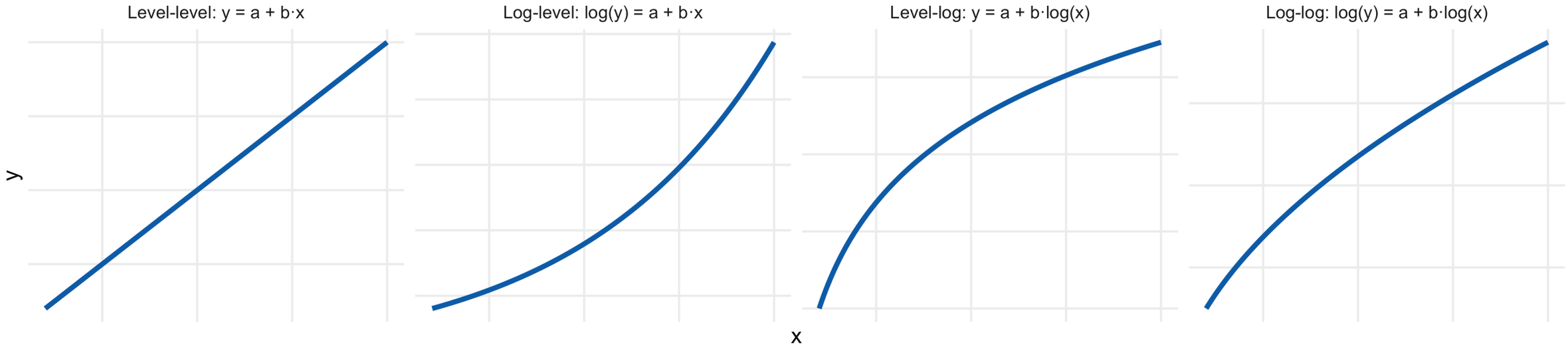
The log-log model: elasticities

```r
model_loglog <- lm(log(streams) ~ log(spend) + log(followers) + placements + genre, data = campaigns)
round(coef(summary(model_loglog)), 3)
```

```
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)          2.881      0.235  12.282    0.000
log(spend)           0.331      0.021  16.001    0.000
log(followers)       0.507      0.016  32.279    0.000
placements           0.120      0.012  10.067    0.000
genreHipHop/Rap      0.199      0.048   4.188    0.000
genreRock           -0.256      0.060  -4.270    0.000
genreElectro/Dance   0.103      0.059   1.728    0.085
```

> `log(spend)` = **0.33**: a 1% increase in spend goes with 0.33% more streams; +10% spend → **+3.3% streams** — the same *percentage* at every budget. Elasticities have no units and are **comparable**: followers (0.51) matter more than the budget.

Four ways to combine levels and logs



Model	Equation	Reading β_1
Level-level	$y = \beta_0 + \beta_1 x$	+1 unit $x \rightarrow \beta_1$ units of y
Log-level	$\log y = \beta_0 + \beta_1 x$	+1 unit $x \rightarrow \approx 100\beta_1$ % of y (exact: $100(e^{\beta_1} - 1)\%$)
Level-log	$y = \beta_0 + \beta_1 \log x$	+1% $x \rightarrow \approx \beta_1/100$ units of y
Log-log	$\log y = \beta_0 + \beta_1 \log x$	+1% $x \rightarrow \approx \beta_1$ % of y (**elasticity**)

Your turn (5 min)

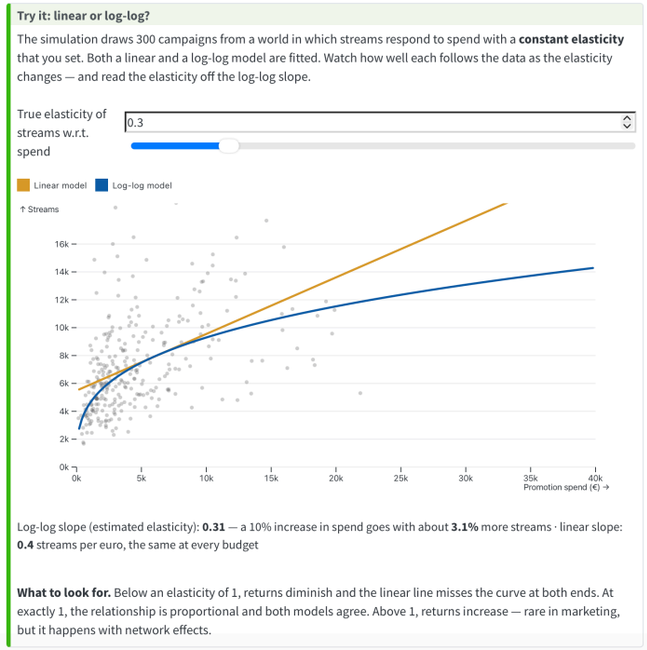
From the log-log model:

1. `genreHipHop/Rap` = 0.20. By what percentage do HipHop/Rap campaigns differ from Pop campaigns (the reference) — approximately and exactly?
2. A marketing manager proposes to double the promotion budget of every campaign. By roughly what percentage would streams increase, according to the model? Is this a causal claim?

Live demo: linear or log-log?

Course book, §5.7 — move the **true elasticity**.

https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-form



> Elasticity **< 1**: diminishing returns, the line misses at both ends. **= 1**: proportional. **> 1**: increasing returns (rare — network effects).

Coding block 3: podcast promotion, revisited (25 min)

Back to `podcast_promo.csv` from coding block 1 — section "Coding block 3" in `05-observational-data.R`.

Laptops open — alone or with your neighbor.

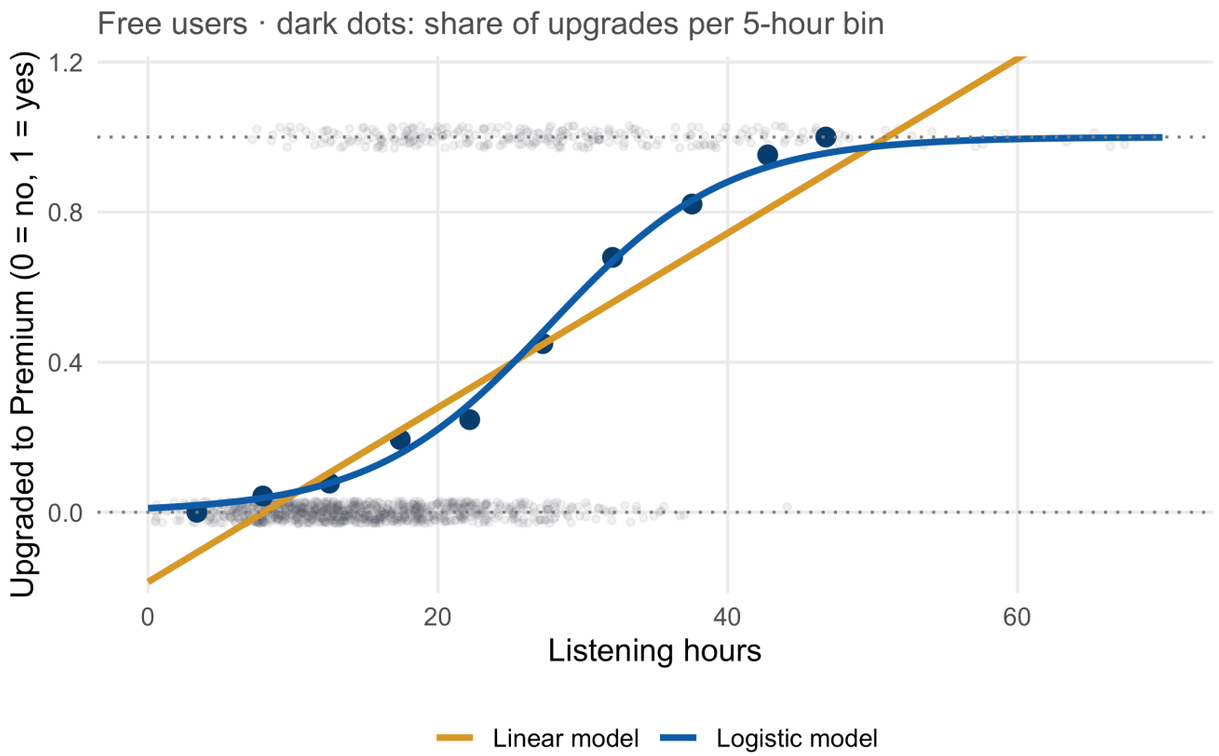
Tasks

1. Scatter plot of downloads against promotion spend with **both axes logged**: what has changed?
2. Fit the **log-log model** on spend and host followers; interpret both **elasticities**: what does +10% spend do?
3. Compare R^2 with the linear model from block 1; which model for the budget advice?

> At the end: one group presents its numbers.

5.8 Yes/no outcomes: logistic regression

Why not a straight line?



- Outcome: **upgraded to Premium** within 3 months — free users, 22% did
- A line predicts probabilities **below 0** and **above 1** — and misses the shape
- The share of upgrades rises slowly, then steeply, then flattens near 100%: an **S-curve**

> We need a function that only takes values between 0 and 1.

The logistic function

$$P(y_i = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

- Very negative $\beta_0 + \beta_1 x_i \rightarrow P \approx 0$ · very positive $\rightarrow P \approx 1$
- Derived from the **log-odds**: odds = $P/(1 - P)$ ("1 to 4" = 0.25)

[Math Processing Error]

- Exponentiate → the odds: $\frac{P(y_i = 1)}{1 - P(y_i = 1)} = e^{\beta_0 + \beta_1 x_i}$

```r
free_users <- filter(smartmix_obs, premium == 0)
model_upgrade <- glm(upgraded ~ hours, family = binomial,
                     data = free_users)
round(coef(summary(model_upgrade)), 3)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   -4.517      0.244  -18.53        0
hours          0.163      0.011   15.05        0
```

> `glm()` = *generalized* linear model: the same predictors, a different link to the outcome. Estimated by **maximum likelihood**, not least squares (no line, no residuals to square): the coefficients under which the observed 0/1 data are most likely.

Reading the coefficients: odds ratios

```
1 exp(coef(model_upgrade))
```

```
(Intercept)       hours
    0.01093     1.17689
```

```
1 exp(confint(model_upgrade))
```

```
              2.5 %  97.5 %
(Intercept) 0.006661 0.01734
hours       1.152949 1.20296
```

- Coefficient 0.163 = change in the **log-odds** per hour — no intuition
- $e^{0.163}$ = **1.18**: each hour multiplies the **odds** of an upgrade by 1.18 (+18%) — the same factor everywhere
- Read **sign** and **significance** directly from the output

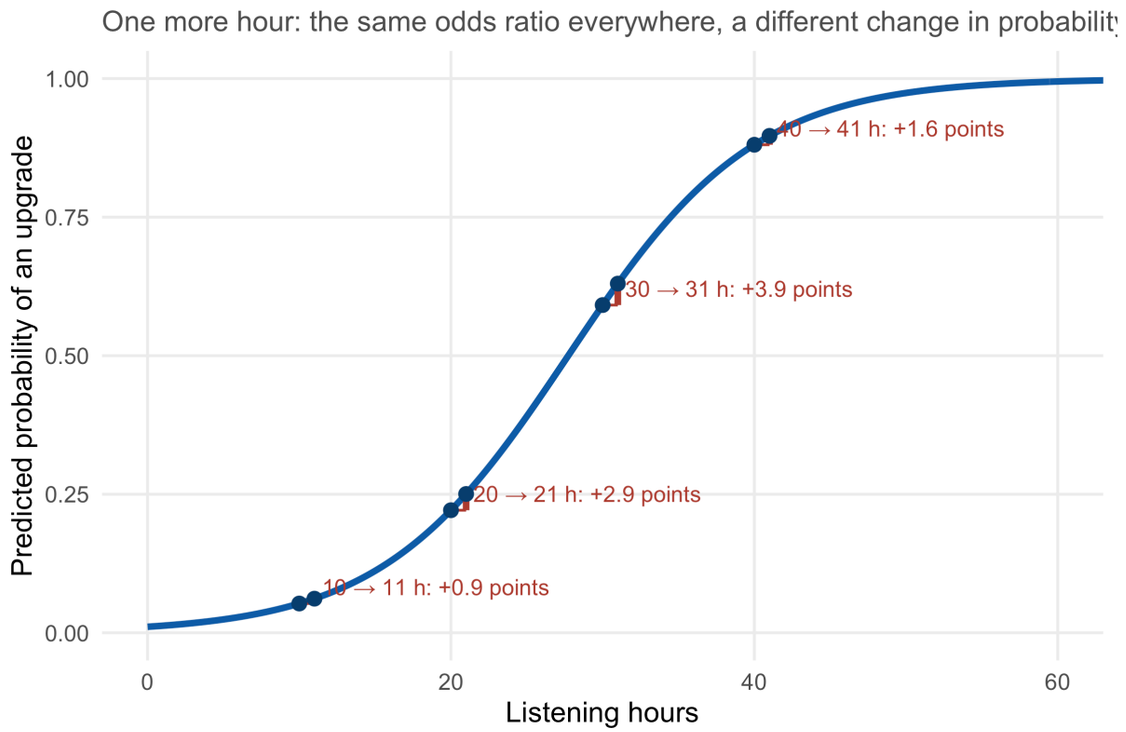
> **Odds ratio ≠ probability ratio.** "Odds +18%" is not "18% more likely" — unless the event is rare. For managers, translate into **probabilities**.

Reading the coefficients: probabilities at x and x + 1

```
predict(model_upgrade, type = "response",
        newdata = data.frame(hours = c(10, 11, 20, 21, 30, 31, 40, 41)))
```

```
      1       2       3       4       5       6       7       8
0.05275 0.06151 0.22111 0.25043 0.59135 0.63005 0.88062 0.89671
```

- 10 → 11 hours: **+0.9** points · 20 → 21: **+2.9** · 30 → 31: **+3.9** · 40 → 41: **+1.6**
- Same coefficient, same odds ratio — a **different change in probability**, largest around 50%



> Report effects as **predicted probabilities at chosen values** (10 vs. 30 hours: 5% vs. 59%) or as the **average change** across all users — “the effect of one more hour” has no single answer in percentage points.

Several predictors: churn

```
model_churn <- glm(churn ~ hours + premium + tenure + smart_mix, family = binomial, data = smartmix_obs)
round(coef(summary(model_churn)), 3)
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)    0.277      0.161   1.717    0.086
hours         -0.061      0.009  -7.130    0.000
premium       -0.533      0.153  -3.489    0.000
tenure        -0.020      0.004  -5.452    0.000
smart_mix     -0.256      0.162  -1.579    0.114
```

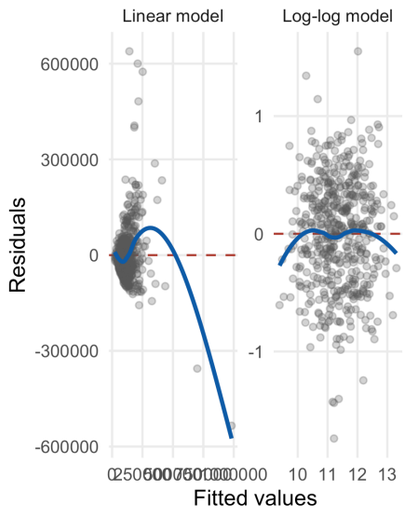
```
p_free    <- predict(model_churn, newdata = mutate(smartmix_obs, premium =
p_premium <- predict(model_churn, newdata = mutate(smartmix_obs, premium =
mean(p_premium - p_free)
```

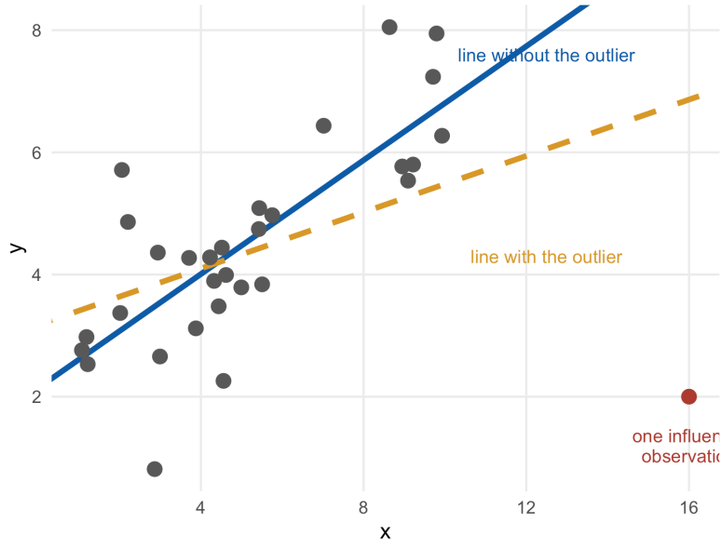
```
[1] -0.06477
```

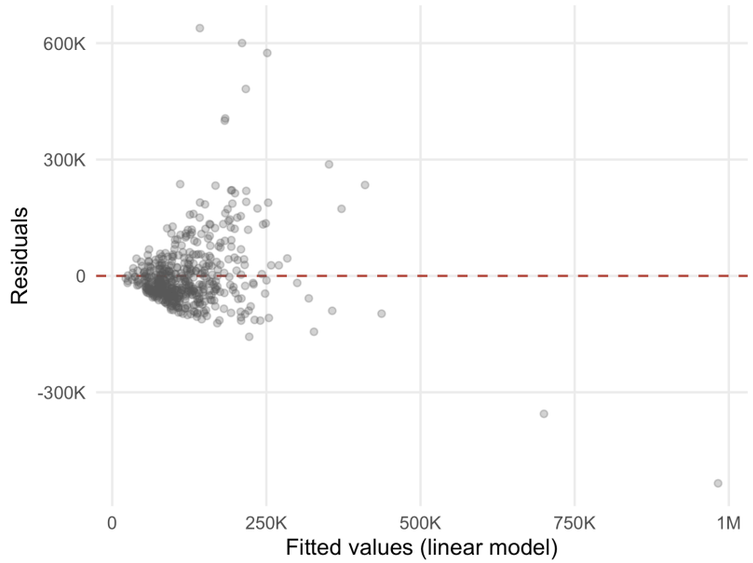
Same logic as multiple regression: each coefficient holds the others constant. Premium subscribers: about **9 percentage points** less likely to churn than otherwise identical free users. Causal? Exogeneity, again.

5.9 The diagnostics that change decisions

Three checks worth doing every time







1. Is the line the right shape? Curved residuals → change the functional form (log) or add a predictor.

2. Does one observation drive the result? `plot(model, which = 5)`; refit without it — if the conclusion changes, say so.

3. Is the uncertainty right? A funnel → **robust standard errors**: `coeftest(model, vcov = vcovHC(model))`.

Coding block 4: winning back users (25 min)

The case. 1,200 users who uninstalled the app got a win-back offer with a **randomly assigned discount** (0–60%). Outcome: reinstalled within 30 days (1/0); also: sessions in the month before uninstalling.

Data: `app_reinstall.csv` — section “Coding block 4” in `05-observational-data.R`

Laptops open — alone or with your neighbor.

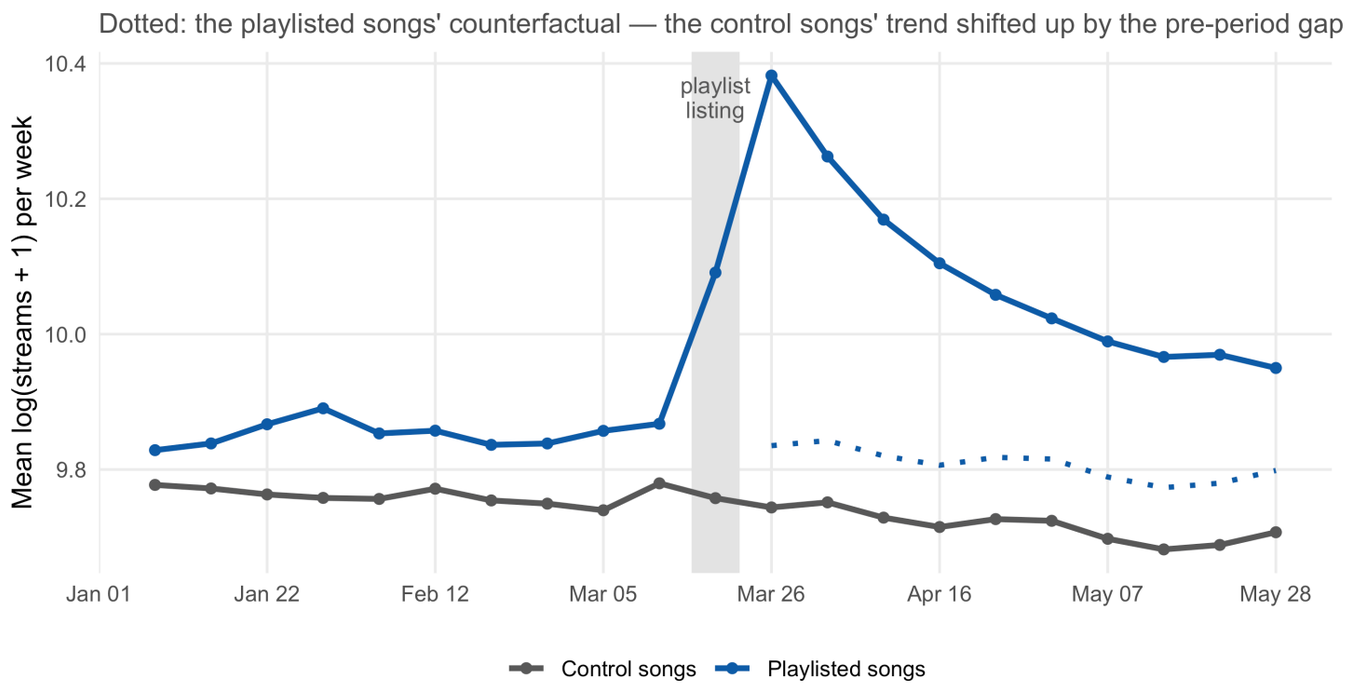
Tasks

1. **Share of reinstalls per discount level**, plotted against the discount: describe the shape
2. **Logistic regression** of `reinstalled` on `discount` and `prior_use`: sign and **odds ratio** of the discount
3. Predict the probability for a user with 10 prior sessions at **10, 11, 40, 41%** discount: how many percentage points per extra point of discount — and why do the two answers differ?

> At the end: one group presents its numbers.

5.10 A quasi-experiment: difference-in-differences

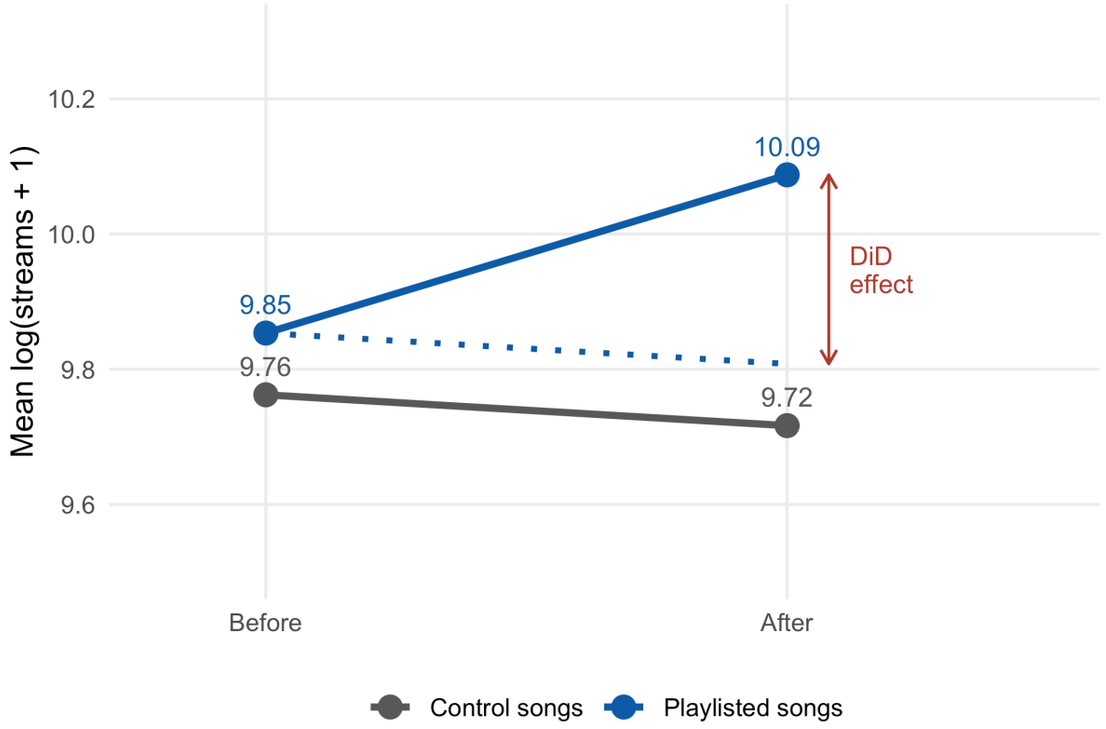
Playlist listings as a natural experiment



- **Real data:** 290 songs × 21 weeks
- March 19, 2018: **53 songs** added to a major playlist; 237 control songs
- Treated songs start **higher** → levels not comparable
- Before the listing: both groups move in **parallel**
- At the listing: the treated group **jumps**

Data: WU-RDS research project (github.com/WU-RDS/did).

Two differences, one estimate



$$\text{DiD} = (\bar{y}_{T,\text{after}} - \bar{y}_{T,\text{before}}) - (\bar{y}_{C,\text{after}} - \bar{y}_{C,\text{before}})$$

$(10.09 - 9.85) - (9.72 - 9.76) =$ **0.28** log points

> The control group's change estimates what would have happened to the treated songs **without** the listing — the counterfactual (dotted).

The DiD regression

```r
did_data <- playlist |>
  filter(week != as.Date("2018-03-19")) |>          # drop the listing week
  mutate(log_streams = log(streams + 1))
model_did <- lm(log_streams ~ treated * post, data = did_data)
round(coef(summary(model_did)), 3)
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)     9.762      0.018 545.865    0.000
treated         0.091      0.042   2.182    0.029
post           -0.046      0.025  -1.803    0.071
treated:post    0.280      0.059   4.727    0.000
```

Coefficient	Meaning
`(Intercept)` 9.76	Control, before: baseline
`treated` 0.09	Treated, before: the **level difference**
`post` −0.05	Control, after − before: the **common time trend**
`treated:post` **0.28**	The extra change of the treated songs: the **DiD effect** → ≈ **+32% streams**

The assumption: parallel trends

```
placebo_data <- did_data |>
  filter(post == 0) |>
  mutate(fake_post = ifelse(week >= as.Date("2018-02-12"), 1, 0))
round(coef(summary(lm(log_streams ~ treated * fake_post,
                      data = placebo_data)))["treated:fake_post", ], 3)
```

```
Estimate Std. Error    t value   Pr(>|t|)
   0.002      0.085      0.027      0.978
```

> Without the listing, the treated songs would have followed the **same trend** as the controls. **Untestable** — but the pre-period makes it plausible: a fake listing in the pre-period shows **no effect**.

Live demo: course book, §5.10

https://wu-rds.github.io/MA2026/05-observational-data.html#sec-obs-did

Your turn (5 min)

Add `major_label` as a control variable to the DiD regression:

```
summary(lm(log_streams ~ treated * post + major_label, data = did_data))
```

Does the effect change? Then: name one situation in which the parallel-trends assumption would **fail** for playlist listings.

What observational data can and cannot tell you

Two strategies, one honest report

- **Adjustment** handles confounders you can **measure** — only as good as your proxies; “enough” is untestable
- **Difference-in-differences** handles confounders **constant over time** — needs parallel trends, plausible but never proven
- **Prediction** needs no causal assumptions — a good predictor is valuable even if no coefficient is an effect

> Report effect + uncertainty + **assumptions**: “Adjusting for prior listening, tenure, Premium and device, adopters listen ≈ 4.5 hours more (95% CI 3.6–5.3), assuming no further differences affect listening — the experiment (3.2) suggests this is not fully met.”

Assignment 3: analyzing observational data

- Your **personal** data (based on your student ID): an opt-in feature, promotion campaigns, and a small panel with a listing
- Tasks: naive vs. adjusted · categorical predictor · elasticity · logistic regression · DiD
- **Deadline:** Group A Sun, Nov 22 · Group B Wed, Nov 25
- **Debrief** at the start of the following session

> **10 points:** report 7 + debrief 3. State your **assumptions** — the best answers say what the data *cannot* show.

Learning check (Part II)

1. Log-log coefficient of ad spend = 0.15. Interpret. The level model says 0.8 sales per euro — why might the two disagree about doubling the budget?
2. Log-level model, dummy “promotion week” = 0.35. Percentage difference in sales — approximate and exact?
3. Residual plot shows a funnel. Consequence for coefficients, for *p*-values — and the simplest fix?
4. A DiD of a price promotion in treated stores: which assumption, and how can the pre-period help?

Next session

Module 6 · Mechanisms & heterogeneity — *Why, and for whom, does it work?*

- Interaction effects: when the effect differs between groups (moderation)
- Mediation: through which mechanism an effect works
- In experiments and observational data — and in your group project